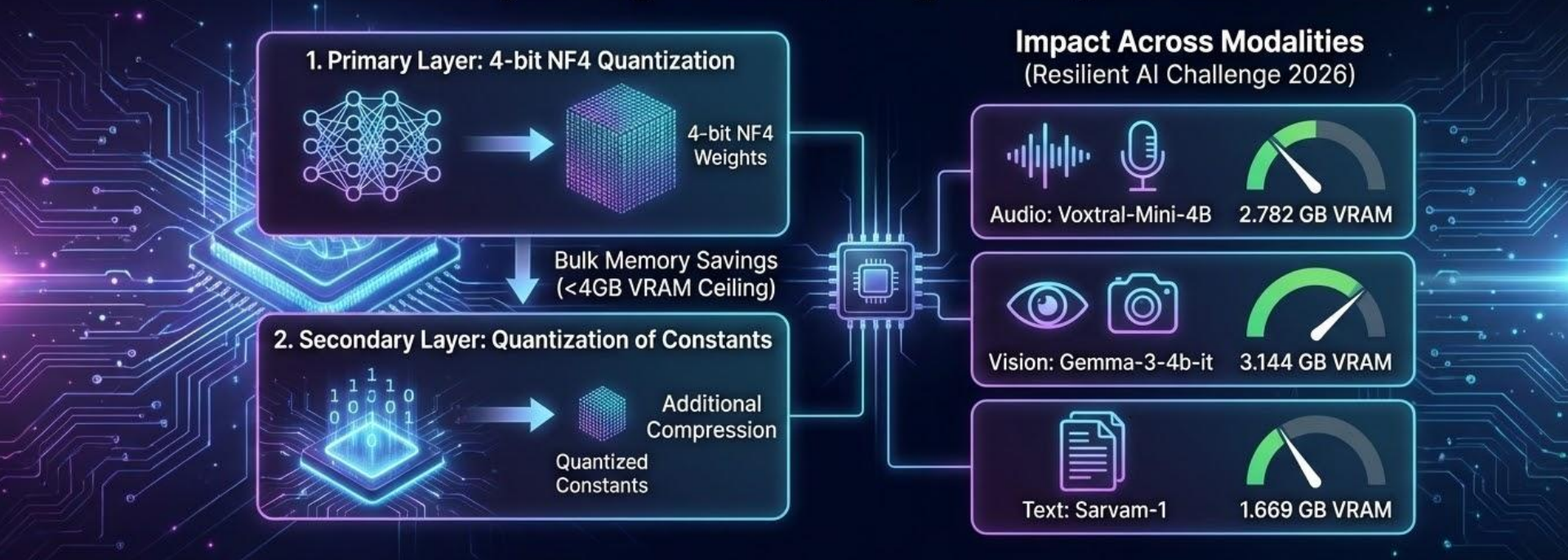
Double Quantization: Reducing AI Memory Footprint

A specialized two-stage compression process for running large AI models on consumer-grade edge hardware, saving ~0.4 bits/parameter.



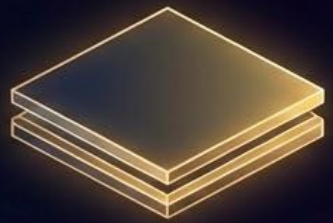
Eliminates unpredictable Memory Spikes and OOM failures, transforming probabilistic AI into a stable, deterministic engineering tool.

Executive Summary: The Double Quantization Mechanism

The strategy employs a two-layer compression stack to fit high-parameter models into a strict <4GB VRAM ceiling:

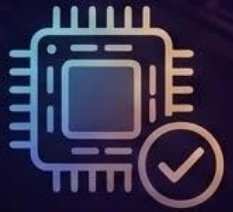
Primary Layer

Uses 4-bit NF4 (NormalFloat 4) quantization to provide the bulk of memory reduction.



Secondary Layer

Quantizes the quantization constants themselves, saving an additional 0.4 bits per parameter.



Core Benefit

This eliminates unpredictable memory spikes and Out-of-Memory (OOM) failures, enabling high-fidelity models to run on consumer-grade edge hardware.

Modality Performance Analysis

The strategy was validated across three distinct modalities for the Resilient AI Challenge 2026, with all models hitting their efficiency targets.

1. Audio Modality (Voxtral-Mini-4B)

The application of double quantization to audio processing focused on maintaining speech recognition fidelity while ensuring real-time response.

VRAM Usage:
2.782 GB (Target <4GB) ✓

Real-Time Factor (RTF):
0.816 (Target <1.0) ✓

Key Insight: By stabilizing the memory footprint, the model could employ dynamic token allocation (70-80 tokens) without risking a system crash during complex audio segments.

Vision Modality (Gemma-3-4b-it)

Vision data is notoriously memory-intensive. The strategy proved that even 4B-parameter vision models can be optimized for the edge.

1. Vision Modality (Gemma-3-4b-it)

The application of double quantization to audio processing focused on maintaining speech recognition fidelity while ensuring real-time response.

VRAM Usage:
3.144 GB (Target <4GB) ✓

Real-Time Factor (RTF):
0.765 (Target <1.0) ✓

Key Insight: The combination of double quantization and Scaled Dot-Product Attention (SDPA) allowed for native 224x224 resolution processing at high speeds.

Text Modality (Sarvam-1)

The text modality achieved the highest level of resource frugality, demonstrating the leanest implementation of the H2E framework.

1. Text Modality (Sarvam-1)

The application of double quantization to audio processing focused on maintaining speech recognition fidelity while ensuring real-time response.

VRAM Usage:
1.669 GB (Target <4GB) ✓

Real-Time Factor (RTF):
0.854 (Target <1.0) ✓

Key Insight: Utilizing reactive response generation (limited to 5 tokens) and greedy decoding ensured that the text output remained 100% deterministic and reproducible.

Future Hardware Integration

The success of this strategy across all three modalities provides the foundation for SOMALA lab's next phase of development:

TPU Optimization

Adapting these quantization protocols for local TPU hardware to maximize battery endurance in 22-DoF humanoid robots.

Power-Aware Control

Implementing loops that adjust inference depth based on remaining battery levels, driven by the predictable power profile of quantized models.

Sovereign Artifacts

Finalizing the transformation of these models into offline, air-gapped artifacts that require zero external API dependencies.